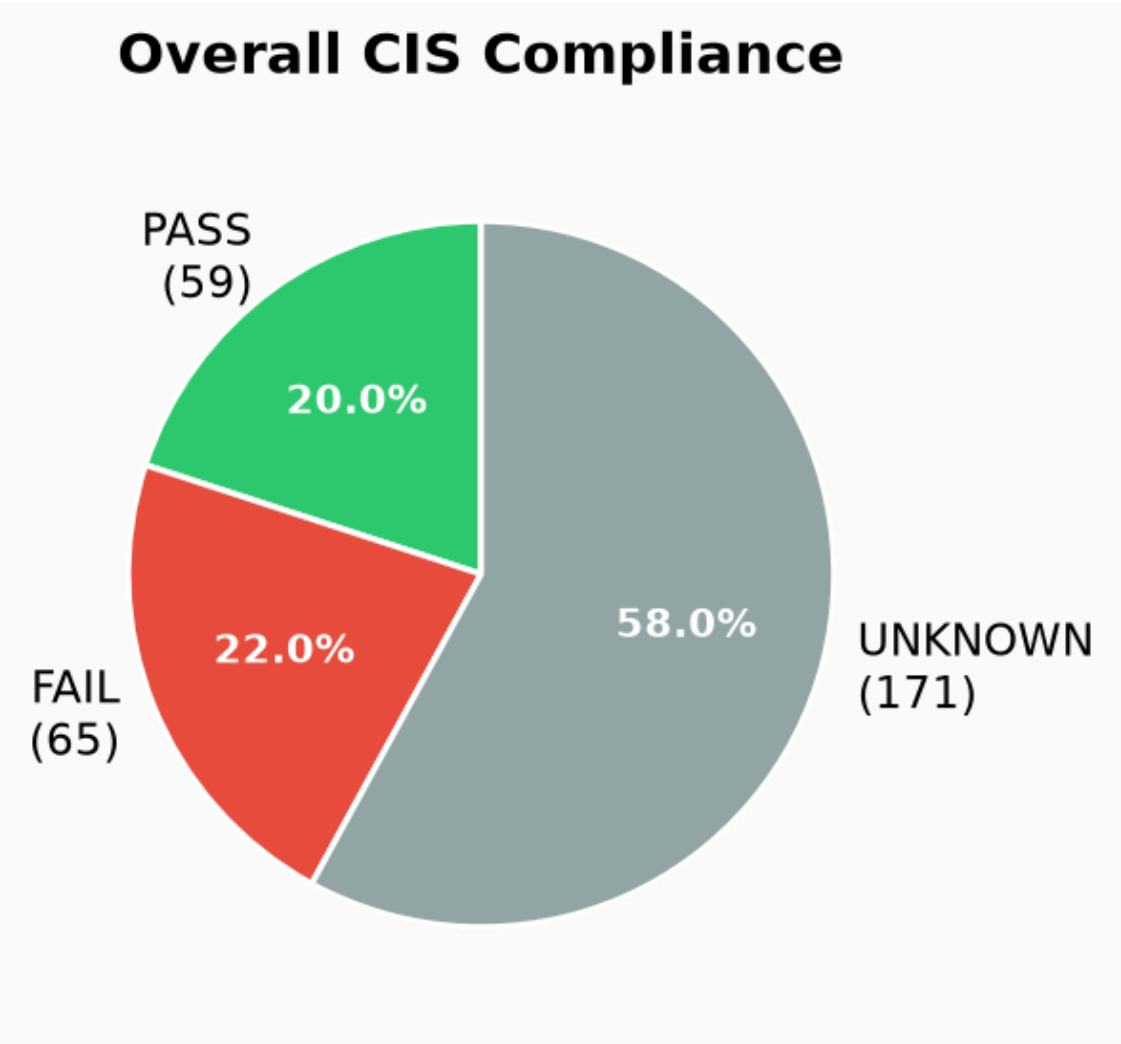


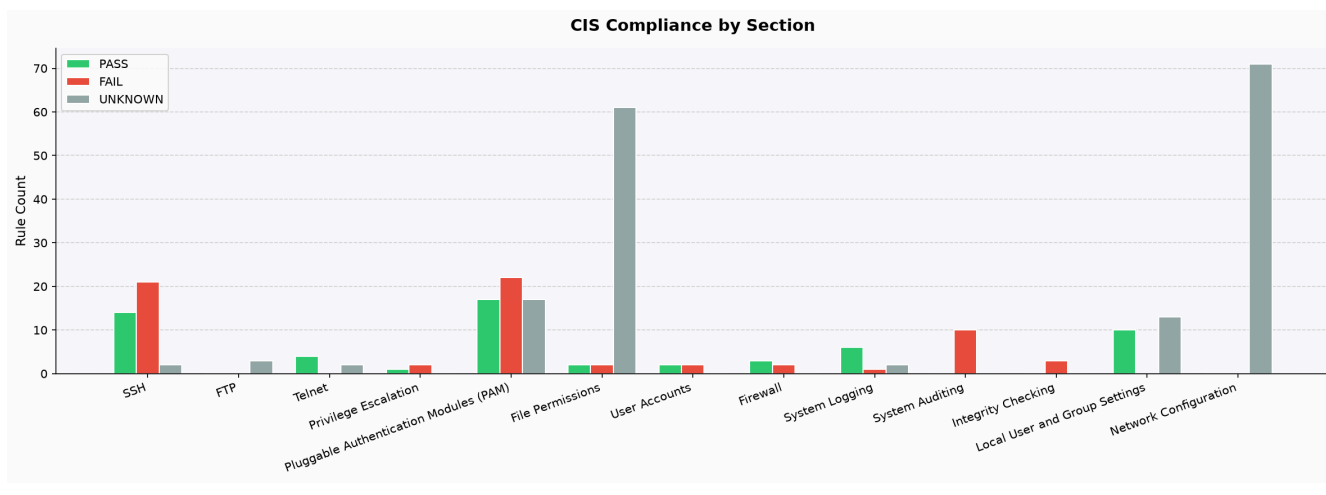
SENTINEL — CIS Benchmark Audit Report

Executive Summary

Metric	Value
Total Rules Evaluated	295
✓ Passed	59
✗ Failed	65
? Unknown	171
Overall Pass Rate	20.0%

Compliance Charts





Per-Section Breakdown

Section	✓ PASS	✗ FAIL	? UNKNOWN	Pass Rate
SSH	14	21	2	38%
FTP	0	0	3	0%
Telnet	4	0	2	67%
Privilege Escalation	1	2	0	33%
Pluggable Authentication Modules (PAM)	17	22	17	30%
File Permissions	2	2	61	3%
User Accounts	2	2	0	50%
Firewall	3	2	0	60%
System Logging	6	1	2	67%
System Auditing	0	10	0	0%
Integrity Checking	0	3	0	0%
Local User and Group Settings	10	0	13	43%
Network Configuration	0	0	71	0%

CIS Benchmark Compliance Report

Here is the security compliance report for the remote machine based on the provided CIS Ubuntu 24.04 LTS Benchmark rules and security profile.

SSH Rules

Rule 1.6.5: Ensure sshd warning Banner is configured

- **Status:** FAIL
- **Evidence:** `ssh.features.banner: "none"`
- **Recommendation:** Configure a warning banner in `/etc/ssh/sshd_config` by setting the `Banner` directive to a file containing the warning message (e.g., `Banner /etc/issue.net`).

Rule 1.6.10: Ensure access to sshd warning banner is configured

- **Status:** INFORMATIONAL
- **Evidence:** `ssh.features.banner: "none"`. No SSH warning banner file is configured.
- **Recommendation:** If a banner is configured (e.g., `Banner /etc/issue.net`), ensure its permissions are 0644 or more restrictive, and it is owned by `root:root`.

Rule 5.1.1: Ensure access to /etc/ssh/sshd_config is configured

- **Status:** FAIL
- **Evidence:** `ssh.sshd_config_permissions: "644 root root /etc/ssh/sshd_config"`
- **Recommendation:** Set permissions of `/etc/ssh/sshd_config` to 0600 or 0640 and ownership to `root:root` using `sudo chmod 0600 /etc/ssh/sshd_config` and `sudo chown root:root /etc/ssh/sshd_config`.

Rule 5.1.2: Ensure access to SSH private host key files is configured

- **Status:** PASS
- **Evidence:** `ssh.ssh_privkey_permissions` shows 0600 root root for all private host key files (`/etc/ssh/ssh_host_ecdsa_key`, `/etc/ssh/ssh_host_ed25519_key`, `/etc/ssh/ssh_host_rsa_key`).
- **Recommendation:** None.

Rule 5.1.3: Ensure access to SSH public host key files is configured

- **Status:** PASS
- **Evidence:** `ssh.ssh_pubkey_permissions` shows 0644 root root for all public host key files (`/etc/ssh/ssh_host_ecdsa_key.pub`, `/etc/ssh/ssh_host_ed25519_key.pub`, `/etc/ssh/ssh_host_rsa_key.pub`).
- **Recommendation:** None.

Rule 5.1.4: Ensure sshd access is configured

- **Status:** FAIL
- **Evidence:** `ssh.authentication.permit_root_login: "without-password"`. No explicit `AllowUsers` or `AllowGroups` directives were found to restrict access.
- **Recommendation:** Configure `AllowUsers` or `AllowGroups` in `/etc/ssh/sshd_config` to explicitly define authorized users or groups for SSH access. Additionally, set `PermitRootLogin no`.

Rule 5.1.5: Ensure sshd Banner is configured

- **Status:** FAIL
- **Evidence:** `ssh.features.banner: "none"`
- **Recommendation:** Configure a warning banner in `/etc/ssh/sshd_config` by setting the `Banner` directive to a file containing the warning message (e.g., `Banner /etc/issue.net`).

Rule 5.1.6: Ensure sshd Ciphers are configured

- **Status:** PASS
- **Evidence:** `ssh.all_ciphers` lists strong ciphers: `chacha20-poly1305@openssh.com`, `aes128-ctr`, `aes192-ctr`, `aes256-ctr`, `aes128-gcm@openssh.com`, `aes256-gcm@openssh.com`.
- **Recommendation:** None.

Rule 5.1.7: Ensure `sshd ClientAliveInterval` and `ClientAliveCountMax` are configured

- **Status:** FAIL
- **Evidence:** `ssh.session.client_alive_interval`: `{"value": 0, "risk": "medium"}`. `ClientAliveInterval` is set to 0.
- **Recommendation:** Set `ClientAliveInterval` to a reasonable non-zero value (e.g., 300 seconds) and `ClientAliveCountMax` to 0-3 in `/etc/ssh/sshd_config` to terminate idle sessions.

Rule 5.1.8: Ensure `sshd DisableForwarding` is enabled

- **Status:** FAIL
- **Evidence:** `ssh.forwarding.allow_tcp_forwarding`: `{"value": true, "risk": "medium"}`, `ssh.forwarding.allow_agent_forwarding`: `{"value": true, "risk": "medium"}`, `ssh.forwarding.allow_streamlocal_forwarding`: `{"value": true, "risk": "medium"}`.
- **Recommendation:** Set `AllowTcpForwarding no`, `AllowAgentForwarding no`, and `AllowStreamLocalForwarding no` in `/etc/ssh/sshd_config` to disable all types of forwarding.

Rule 5.1.9: Ensure `sshd GSSAPIAuthentication` is disabled

- **Status:** PASS
- **Evidence:** `ssh.authentication.gssapiauthentication`: `false`
- **Recommendation:** None.

Rule 5.1.10: Ensure `sshd HostbasedAuthentication` is disabled

- **Status:** PASS
- **Evidence:** `ssh.authentication.hostbasedauthentication`: `false`
- **Recommendation:** None.

Rule 5.1.11: Ensure `sshd IgnoreRhosts` is enabled

- **Status:** PASS
- **Evidence:** `ssh.authentication.ignorerhosts`: `true`
- **Recommendation:** None.

Rule 5.1.12: Ensure `sshd KexAlgorithms` is configured

- **Status:** FAIL
- **Evidence:** `crypto.weak_kex` lists multiple weak algorithms: `ecdh-sha2-nistp256`, `ecdh-sha2-nistp384`, `ecdh-sha2-nistp521`, `diffie-hellman-group14-sha256`.

- **Recommendation:** Configure KexAlgorithms in `/etc/ssh/sshd_config` to use only strong key exchange algorithms, excluding those listed as weak. Refer to the CIS benchmark for a recommended list.

Rule 5.1.13: Ensure sshd LoginGraceTime is configured

- **Status:** FAIL
- **Evidence:** `ssh.session.login_grace_time: 120`
- **Recommendation:** Set LoginGraceTime to 60 seconds or less in `/etc/ssh/sshd_config`.

Rule 5.1.14: Ensure sshd LogLevel is configured

- **Status:** FAIL
- **Evidence:** `ssh.features.log_level: "INFO"`
- **Recommendation:** Set LogLevel VERBOSE in `/etc/ssh/sshd_config` for enhanced logging and auditing.

Rule 5.1.15: Ensure sshd MACs are configured

- **Status:** FAIL
- **Evidence:** `crypto.weak_macs` lists multiple weak algorithms, including `hmac-sha1-etm@openssh.com` which is vulnerable to the Terrapin attack (CVE-2023-48795).
- **Recommendation:** Configure MACs in `/etc/ssh/sshd_config` to use only strong Message Authentication Codes, excluding those listed as weak. Refer to the CIS benchmark for a recommended list.

Rule 5.1.16: Ensure sshd MaxAuthTries is configured

- **Status:** FAIL
- **Evidence:** `ssh.authentication.max_auth_tries: 6`
- **Recommendation:** Set MaxAuthTries to 4 or less in `/etc/ssh/sshd_config`.

Rule 5.1.17: Ensure sshd MaxStartups is configured

- **Status:** FAIL
- **Evidence:** `ssh.session.maxstartups: "10:30:100"`
- **Recommendation:** Set MaxStartups to 10:30:60 or less in `/etc/ssh/sshd_config`.

Rule 5.1.18: Ensure sshd MaxSessions is configured

- **Status:** FAIL
- **Evidence:** `ssh.session.max_sessions: 10`
- **Recommendation:** Set MaxSessions to 4 or less in `/etc/ssh/sshd_config`.

Rule 5.1.19: Ensure sshd PermitEmptyPasswords is disabled

- **Status:** PASS
- **Evidence:** `ssh.authentication.permit_empty_passwords: false`
- **Recommendation:** None.

Rule 5.1.20: Ensure sshd PermitRootLogin is disabled

- **Status:** FAIL
- **Evidence:** `ssh.authentication.permit_root_login: "without-password"`
- **Recommendation:** Set `PermitRootLogin no` in `/etc/ssh/sshd_config` to completely disable direct root login via SSH.

Rule 5.1.21: Ensure sshd PermitUserEnvironment is disabled

- **Status:** PASS
- **Evidence:** `ssh.features.permit_user_environment: false`
- **Recommendation:** None.

Rule 5.1.22: Ensure sshd UsePAM is enabled

- **Status:** PASS
- **Evidence:** `ssh.authentication.usepam: true`
- **Recommendation:** None.

Rule 5.1.23: Ensure sshd post-quantum cryptography key exchange algorithms are configured

- **Status:** UNKNOWN
- **Evidence:** The profile does not explicitly list post-quantum cryptography key exchange algorithms. `crypto.weak_kex` lists several weak algorithms, suggesting PQC KEX are not optimally configured.
- **Recommendation:** Review and configure `KexAlgorithms` in `/etc/ssh/sshd_config` to include post-quantum safe algorithms as recommended by CIS.

Rule 5.1.24: Ensure sshd ListenAddress is configured

- **Status:** FAIL
- **Evidence:** `ssh.network.listen_address: "0.0.0.0:22"`
- **Recommendation:** Set `ListenAddress` to a specific IP address (e.g., `192.168.1.10:22`) in `/etc/ssh/sshd_config` to restrict the SSH daemon to a single network interface.

FTP Rules

Rule 2.1.8: Ensure ftp server services are not in use

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `services` data could not be collected. Ensure no FTP server packages are installed and no FTP server units are active.

Rule 2.1.20: Ensure tftp server services are not in use

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile reported an error: No JSON

object found in output: ''.

- **Recommendation:** Investigate why `services` data could not be collected. Ensure no TFTP server packages are installed and no TFTP server units are active.

Rule 2.2.6: Ensure ftp client is not installed

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `services` data could not be collected. Ensure no FTP client packages are installed.

TELNET Rules

Rule 2.1.19: Ensure telnet server services are not in use

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `services` data could not be collected. Ensure no Telnet server packages are installed and no Telnet server units are active.

Rule 2.2.4: Ensure telnet client is not installed

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `services` data could not be collected. Ensure no Telnet client packages are installed.

ESCALATION Rules

Rule 5.2.1: Ensure sudo is installed

- **Status:** PASS
- **Evidence:** `privilege_escalation.sudo_installed.installed: true`
- **Recommendation:** None.

Rule 5.2.2: Ensure sudo commands use pty

- **Status:** PASS
- **Evidence:** `privilege_escalation.use_pty_entries` contains `{"file": "/etc/sudoers", "content": "Defaults\tuse_pty"}`.
- **Recommendation:** None.

Rule 5.2.3: Ensure sudo log file exists

- **Status:** PASS
- **Evidence:** `privilege_escalation.sudo_syslog_logging.auth_log_exists: true, syslog_log_exists: true, and rsyslog_authpriv_configured: true.`

- **Recommendation:** None.

Rule 5.2.4: Ensure users must provide password for escalation

- **Status:** PASS
- **Evidence:** `privilege_escalation.nopasswd_entries` is empty, indicating no NOPASSWD directives are configured.
- **Recommendation:** None.

Rule 5.2.5: Ensure re-authentication for privilege escalation is not disabled globally

- **Status:** PASS
- **Evidence:** `privilege_escalation.noauthenticate_entries` is empty.
- **Recommendation:** None.

Rule 5.2.6: Ensure sudo timestamp_timeout is configured

- **Status:** FAIL
- **Evidence:** `privilege_escalation.timestamp_timeout_entries` is empty, indicating `timestamp_timeout` is not explicitly configured.
- **Recommendation:** Add `Defaults timestamp_timeout=0` or `Defaults timestamp_timeout=15` to `/etc/sudoers` or a file in `/etc/sudoers.d/` to enforce re-authentication or a short timeout.

Rule 5.2.7: Ensure access to the su command is restricted

- **Status:** FAIL
- **Evidence:** The `pam_su_config` shows `pam_wheel.so` is commented out (`# auth required pam_wheel.so`), meaning `su` access is not restricted to a specific group.
- **Recommendation:** Uncomment `auth required pam_wheel.so` in `/etc/pam.d/su` and ensure only authorized users are members of the `wheel` group.

DIRECTORY ACCESS Rules

Rule 7.1.1: Ensure access to /etc/passwd is configured

- **Status:** PASS
- **Evidence:** `/etc/passwd` has permissions `0644` and is owned by `root:root`.
- **Recommendation:** None.

Rule 7.1.2: Ensure access to /etc/passwd- is configured

- **Status:** PASS
- **Evidence:** `/etc/passwd-` has permissions `0644` and is owned by `root:root`.
- **Recommendation:** None.

Rule 7.1.3: Ensure access to /etc/group is configured

- **Status:** PASS
- **Evidence:** `/etc/group` has permissions `0644` and is owned by `root:root`.

- **Recommendation:** None.

Rule 7.1.4: Ensure access to /etc/group- is configured

- **Status:** PASS
- **Evidence:** /etc/group- has permissions 0644 and is owned by root:root.
- **Recommendation:** None.

Rule 7.1.5: Ensure access to /etc/shadow is configured

- **Status:** PASS
- **Evidence:** /etc/shadow has permissions 0640 and is owned by root:shadow.
- **Recommendation:** None.

Rule 7.1.6: Ensure access to /etc/shadow- is configured

- **Status:** PASS
- **Evidence:** /etc/shadow- has permissions 0640 and is owned by root:shadow.
- **Recommendation:** None.

Rule 7.1.7: Ensure access to /etc/gshadow is configured

- **Status:** PASS
- **Evidence:** /etc/gshadow has permissions 0640 and is owned by root:shadow.
- **Recommendation:** None.

Rule 7.1.8: Ensure access to /etc/gshadow- is configured

- **Status:** PASS
- **Evidence:** /etc/gshadow- has permissions 0640 and is owned by root:shadow.
- **Recommendation:** None.

Rule 7.1.9: Ensure access to /etc/shells is configured

- **Status:** PASS
- **Evidence:** /etc/shells has permissions 0644 and is owned by root:root.
- **Recommendation:** None.

Rule 7.1.10: Ensure access to /etc/security/opasswd is configured

- **Status:** PASS
- **Evidence:** /etc/security/opasswd has permissions 0600 and is owned by root:root.
- **Recommendation:** None.

Rule 7.1.11: Ensure world writable files and directories are secured

- **Status:** PASS
- **Evidence:** file_permissions.world_writable.world_writable_files and file_permissions.world_writable.world_writable_dirs_no_sticky are empty for scanned mounts.
- **Recommendation:** None.

Rule 7.1.12: Ensure no files or directories without an owner and a group exist

- **Status:** PASS
- **Evidence:** `file_permissions.unowned.unowned_or_ungrouped` is empty for scanned mounts.
- **Recommendation:** None.

Rule 7.1.13: Ensure SUID and SGID files are reviewed

- **Status:** INFORMATIONAL
- **Evidence:** `file_permissions.suid_sgid.suid_sgid_files` lists SUID/SGID files such as `/usr/bin/umount`, `/usr/bin/sudo`, `/usr/bin/passwd`, `/usr/bin/chage`, etc.
- **Recommendation:** Manually review the listed SUID/SGID files to ensure they are authorized and necessary for system operation.

USER ACCOUNTS Rules

Rule 5.4.1.1: Ensure password expiration is configured

- **Status:** FAIL
- **Evidence:** `PASS_MAX_DAYS` is set to 99999 in `/etc/login.defs` and for most users, meaning passwords never expire.
- **Recommendation:** Set `PASS_MAX_DAYS` to 365 or less (e.g., 90) in `/etc/login.defs` and apply to existing users using `chage -M <days> <username>`.

Rule 5.4.1.2: Ensure minimum password days is configured

- **Status:** FAIL
- **Evidence:** `PASS_MIN_DAYS` is set to 0 in `/etc/login.defs` and for most users, meaning users can change passwords immediately.
- **Recommendation:** Set `PASS_MIN_DAYS` to 1 or more (e.g., 7) in `/etc/login.defs` and apply to existing users using `chage -m <days> <username>`.

Rule 5.4.1.3: Ensure password expiration warning days is configured

- **Status:** PASS
- **Evidence:** `PASS_WARN_AGE` is set to 7 in `/etc/login.defs` and for most users.
- **Recommendation:** None.

Rule 5.4.1.4: Ensure strong password hashing algorithm is configured

- **Status:** PASS
- **Evidence:** `ENCRYPT_METHOD` is SHA512 in `/etc/login.defs` and `yescrypt` is present in `pam_unix` configuration.
- **Recommendation:** None.

Rule 5.4.1.5: Ensure inactive password lock is configured

- **Status:** FAIL

- **Evidence:** `useradd_inactive` is `-1` and `account_inactive` is never for most users, meaning accounts are never locked due to inactivity.
- **Recommendation:** Set `INACTIVE` to 30 or less (e.g., 30) in `/etc/default/useradd` and apply to existing users using `chage -I <days> <username>`.

Rule 5.4.1.6: Ensure all users last password change date is in the past

- **Status:** FAIL
- **Evidence:** Multiple users (e.g., `root`, `sebti`, `sshd`) have `last_password_change` dates in the future (e.g., "Feb 16, 2025", "Apr 19, 2026", "Jun 25, 2026").
- **Recommendation:** Force all users to change their passwords immediately using `chage -d 0 <username>` or ensure their last password change date is in the past.

Rule 5.4.2.1: Ensure root is the only UID 0 account

- **Status:** PASS
- **Evidence:** `uid_0_accounts` lists only "root".
- **Recommendation:** None.

Rule 5.4.2.2: Ensure root is the only GID 0 account

- **Status:** PASS
- **Evidence:** `primary_gid_0_accounts` lists only "root".
- **Recommendation:** None.

Rule 5.4.2.3: Ensure group root is the only GID 0 group

- **Status:** PASS
- **Evidence:** `gid_0_groups` lists only "root".
- **Recommendation:** None.

Rule 5.4.2.4: Ensure root account access is controlled

- **Status:** FAIL
- **Evidence:** `securetty_exists: false`. `PermitRootLogin` is "without-password" (see 5.1.20).
- **Recommendation:** Create `/etc/securetty` and list allowed TTYs for root login. Set `PermitRootLogin no` in `/etc/ssh/sshd_config`.

Rule 5.4.2.5: Ensure root path integrity

- **Status:** FAIL
- **Evidence:** `root_path` includes `/usr/games`, `/usr/local/games`, and `/snap/bin`, which are not typically considered secure for root's `PATH`.
- **Recommendation:** Ensure root's `PATH` environment variable is restricted to only necessary and secure directories (e.g., `/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin`).

Rule 5.4.2.6: Ensure root user umask is configured

- **Status:** FAIL
- **Evidence:** `root_umask` from `login_defs` is `022`.

- **Recommendation:** Set root's umask to 077 in `/root/.bashrc` and `/root/.profile`, and ensure `UMASK` in `/etc/login.defs` is 077.

Rule 5.4.2.7: Ensure system accounts do not have a valid login shell

- **Status:** FAIL
- **Evidence:** The system account `sync` has `/bin/sync` as its shell, which is a valid login shell.
- **Recommendation:** Change the shell for the `sync` user to `/usr/sbin/nologin` or `/bin/false` using `usermod -s /usr/sbin/nologin sync`.

Rule 5.4.2.8: Ensure accounts without a valid login shell are locked

- **Status:** PASS
- **Evidence:** All listed system accounts have `passwd_status: "L"`, indicating they are locked.
- **Recommendation:** None.

Rule 5.4.3.1: Ensure nologin is not listed in /etc/shells

- **Status:** PASS
- **Evidence:** `nologin_in_shells: false`.
- **Recommendation:** None.

Rule 5.4.3.2: Ensure default user shell timeout is configured

- **Status:** FAIL
- **Evidence:** No `TMOUT` settings found in `/etc/profile`, `/etc/profile.d`, or `/etc/bash.bashrc`.
- **Recommendation:** Add `TMOUT=600` (or less) to `/etc/profile.d/timeout.sh` to automatically log out idle shell sessions.

Rule 5.4.3.3: Ensure default user umask is configured

- **Status:** FAIL
- **Evidence:** `default_user_umask.login_defs` is 022.
- **Recommendation:** Set `UMASK` to 027 or 077 in `/etc/login.defs` and ensure `pam_umask.so` is configured to enforce it.

FIREWALL Rules

Rule 4.1.1: Ensure ufw is installed

- **Status:** PASS
- **Evidence:** `ufw_firewall.ufw_installed: true`
- **Recommendation:** None.

Rule 4.1.2: Ensure ufw service is configured

- **Status:** FAIL
- **Evidence:** `ufw_firewall.service_configured.is_enabled: "enabled"`,

`is_active: "active"`, but `status_verbose: "Status: inactive"`. This indicates the UFW service is enabled but the firewall itself is inactive.

- **Recommendation:** Ensure UFW is enabled and active by running `sudo ufw enable`.

Rule 4.1.3: Ensure ufw incoming default is configured

- **Status:** PASS
- **Evidence:** `DEFAULT_INPUT_POLICY` is `DROP` in `/etc/default/ufw`.
- **Recommendation:** None.

Rule 4.1.4: Ensure ufw outgoing default is configured

- **Status:** FAIL
- **Evidence:** `DEFAULT_OUTPUT_POLICY` is `ACCEPT` in `/etc/default/ufw`.
- **Recommendation:** Set `DEFAULT_OUTPUT_POLICY` to `DROP` in `/etc/default/ufw` and configure explicit allow rules for necessary outgoing traffic.

Rule 4.1.5: Ensure ufw routed default is configured

- **Status:** PASS
- **Evidence:** `DEFAULT_FORWARD_POLICY` is `DROP` in `/etc/default/ufw`.
- **Recommendation:** None.

SYSTEM LOGGING Rules

Rule 6.1.1.1.1: Ensure systemd-journald service is enabled and active

- **Status:** PASS
- **Evidence:** `systemd-journald_enabled: "static"` and `systemd-journald_active: "active"`.
- **Recommendation:** None.

Rule 6.1.1.1.2: Ensure systemd-journal-remote service is not in use

- **Status:** PASS
- **Evidence:** `systemd-journal-remote_installed: false`.
- **Recommendation:** None.

Rule 6.1.1.1.3: Ensure journald is configured to send logs to rsyslog

- **Status:** FAIL
- **Evidence:** `ForwardToSyslog=no` is commented out in `/etc/systemd/journald.conf`, implying the default (no forwarding) is in effect.
- **Recommendation:** Uncomment and set `ForwardToSyslog=yes` in `/etc/systemd/journald.conf` to ensure logs are sent to rsyslog.

Rule 6.1.1.1.4: Ensure journald log file access is configured

- **Status:** INFORMATIONAL
- **Evidence:** Directories `/run/log/journal` and `/var/log/journal` have `02755` permissions, owned by `root:systemd-journal`. Log files within `/var/log/`

journal have 0640 permissions, owned by root:systemd-journal.

- **Recommendation:** Manually verify that these permissions are appropriate for the environment and that the systemd-journal group has only authorized members.

Rule 6.1.1.1.5: Ensure journald log file rotation is configured

- **Status:** INFORMATIONAL
- **Evidence:** Log rotation settings like SystemMaxUse, SystemKeepFree, SystemMaxFileSize, SystemMaxFiles are commented out in /etc/systemd/journald.conf, indicating default values are used.
- **Recommendation:** Manually review the default journald log rotation settings and configure SystemMaxUse, SystemKeepFree, SystemMaxFileSize, or SystemMaxFiles in /etc/systemd/journald.conf to meet retention requirements.

Rule 6.1.1.1.6: Ensure journald Storage is configured

- **Status:** PASS
- **Evidence:** Storage=auto is commented out, but /var/log/journal exists, implying persistent storage is active.
- **Recommendation:** None (or explicitly set Storage=persistent for clarity).

Rule 6.1.1.1.7: Ensure journald Compress is configured

- **Status:** PASS
- **Evidence:** Compress=yes is commented out, implying the default (compression enabled) is in effect.
- **Recommendation:** None (or explicitly set Compress=yes for clarity).

Rule 6.1.2.1: Ensure rsyslog is installed

- **Status:** PASS
- **Evidence:** rsyslog_installed: true.
- **Recommendation:** None.

Rule 6.1.2.2: Ensure rsyslog service is enabled and active

- **Status:** PASS
- **Evidence:** rsyslog_enabled: "enabled" and rsyslog_active: "active".
- **Recommendation:** None.

Rule 6.1.2.3: Ensure rsyslog default file permissions are configured

- **Status:** PASS
- **Evidence:** \$FileCreateMode 0640, \$DirCreateMode 0755, \$Umask 0022 are configured in /etc/rsyslog.conf.
- **Recommendation:** None.

Rule 6.1.2.4: Ensure rsyslog logging is configured

- **Status:** INFORMATIONAL
- **Evidence:** rsyslog_configs shows default logging rules in /etc/rsyslog.d/50-

default.conf (e.g., auth, authpriv.* /var/log/auth.log).

- **Recommendation:** Manually review /etc/rsyslog.conf and files in /etc/rsyslog.d/ to ensure all required system events are logged according to organizational policy.

Rule 6.1.2.5: Ensure rsyslog is configured to send logs to a remote log host

- **Status:** INFORMATIONAL
- **Evidence:** No remote log host forwarding configuration found in rsyslog_configs.
- **Recommendation:** Manually verify if remote logging is required. If so, configure rsyslog to send logs to a remote log host.

Rule 6.1.2.6: Ensure rsyslog is not configured to receive logs from a remote client

- **Status:** PASS
- **Evidence:** imudp and imtcp modules are commented out in /etc/rsyslog.conf, indicating remote reception is disabled.
- **Recommendation:** None.

Rule 6.1.2.7: Ensure logrotate is configured

- **Status:** INFORMATIONAL
- **Evidence:** logrotate is installed and active, with configurations in /etc/logrotate.conf and /etc/logrotate.d/.
- **Recommendation:** Manually review logrotate configurations to ensure all critical log files are rotated, compressed, and retained according to policy.

Rule 6.1.2.8: Ensure rsyslog-gnutls is installed

- **Status:** FAIL
- **Evidence:** rsyslog_gnutls_installed: false.
- **Recommendation:** Install the rsyslog-gnutls package to enable secure remote log forwarding.

Rule 6.1.2.9: Ensure rsyslog is configured to use gtls for forwarding

- **Status:** FAIL
- **Evidence:** rsyslog_gnutls_installed: false, therefore gtls cannot be configured for forwarding.
- **Recommendation:** Install rsyslog-gnutls and configure rsyslog to use gtls for any remote log forwarding.

Rule 6.1.2.10: Ensure rsyslog CA certificates are configured

- **Status:** INFORMATIONAL
- **Evidence:** rsyslog_gnutls_installed: false. CA certificates are not applicable without gnutls.
- **Recommendation:** If rsyslog-gnutls is installed and remote forwarding is configured, ensure CA certificates are properly configured for secure communication.

Rule 6.1.3.1: Ensure access to all logfiles has been configured

- **Status:** INFORMATIONAL
- **Evidence:** `system_logging.logfiles` lists permissions for various log files (e.g., `/var/log/auth.log` is `0640 root adm`, `/var/log/syslog` is `0640 syslog adm`).
- **Recommendation:** Manually review the permissions of all log files to ensure they are appropriately restrictive and owned by authorized users/groups.

SYSTEM AUDITING (auditd) Rules

Rule 6.2.1.1.1: Ensure auditd is installed

- **Status:** FAIL
- **Evidence:** `auditd_installed: false`.
- **Recommendation:** Install the auditd package using `sudo apt install auditd`.

Rule 6.2.1.1.2: Ensure auditd service is enabled and active

- **Status:** FAIL
- **Evidence:** `auditd_installed: false`.
- **Recommendation:** Install auditd and then enable and start the auditd service using `sudo systemctl enable auditd` and `sudo systemctl start auditd`.

Rule 6.2.1.1.3: Ensure auditing for processes that start prior to auditd is enabled

- **Status:** FAIL
- **Evidence:** `auditd_installed: false`.
- **Recommendation:** Install auditd and configure GRUB to include `audit=1` in the kernel command line.

Rule 6.2.1.1.4: Ensure audit_backlog_limit is sufficient

- **Status:** FAIL
- **Evidence:** `auditd_installed: false`.
- **Recommendation:** Install auditd and configure `audit_backlog_limit` in `/etc/audit/audit.conf` to a sufficient value (e.g., 8192).

Rule 6.2.1.2.1: Ensure audit log storage size is configured

- **Status:** FAIL
- **Evidence:** `auditd_installed: false`.
- **Recommendation:** Install auditd and configure `max_log_file` in `/etc/audit/auditd.conf` to an appropriate size (e.g., `max_log_file = 50`).

Rule 6.2.1.2.2: Ensure audit logs are not automatically deleted

- **Status:** FAIL
- **Evidence:** `auditd_installed: false`.
- **Recommendation:** Install auditd and set `max_log_file_action = keep_logs` in

`/etc/audit/auditd.conf.`

Rule 6.2.1.2.3: Ensure system is disabled when audit logs are full

- **Status:** FAIL
- **Evidence:** `auditd_installed: false.`
- **Recommendation:** Install `auditd` and set `space_left_action = SYSLOG` and `admin_space_left_action = HALT` in `/etc/audit/auditd.conf`.

Rule 6.2.1.2.4: Ensure system warns when audit logs are low on space

- **Status:** FAIL
- **Evidence:** `auditd_installed: false.`
- **Recommendation:** Install `auditd` and set `space_left_action = SYSLOG` in `/etc/audit/auditd.conf`.

Rule 6.2.3.1-6.2.3.30: Ensure auditd rules cover sudoers, identity, logins, session, time-change, MAC policy, access, mounts, deletion, loading modules, and configuration is immutable

- **Status:** FAIL
- **Evidence:** `auditd_installed: false.`
- **Recommendation:** Install `auditd` and configure comprehensive audit rules in `/etc/audit/rules.d/audit.rules` as per CIS benchmark. Ensure the rules are made immutable by adding `-e 2` to the end of the rules file.

Rule 6.2.4.1-6.2.4.10: Ensure audit log files and audit tools file access and permissions are strictly constrained

- **Status:** FAIL
- **Evidence:** `auditd_installed: false.`
- **Recommendation:** Install `auditd` and ensure audit log files (e.g., `/var/log/audit/audit.log`) and audit tools (e.g., `/sbin/auditctl`) have restrictive permissions (e.g., `0640` for logs, `0755` for tools) and are owned by `root:root`.

Integrity Checking Rules

Rule 6.3.1: Ensure AIDE is installed

- **Status:** FAIL
- **Evidence:** `aide_integrity_checking.aide_installed: false.`
- **Recommendation:** Install the `aide` package using `sudo apt install aide`.

Rule 6.3.2: Ensure filesystem integrity is regularly checked

- **Status:** FAIL
- **Evidence:** `aide_integrity_checking.aide_installed: false.`
- **Recommendation:** Install `aide` and configure a cron job or systemd timer to regularly run `aide --check`.

Rule 6.3.3: Ensure cryptographic mechanisms protect tools

- **Status:** FAIL
- **Evidence:** `aide_integrity_checking.aide_installed: false.`
- **Recommendation:** Install `aide` and configure its rules in `/etc/aide/aide.conf` to include cryptographic hashes (e.g., `sha512`) for critical system binaries and configuration files.

LOCAL USER Rules

Rule 7.2.1: Ensure accounts in `/etc/passwd` use shadowed passwords

- **Status:** PASS
- **Evidence:** `/etc/shadow` exists and is properly configured (7.1.5). User password statuses in `user_accounts.root_and_system_accounts.system_accounts` are `L` or `*`.
- **Recommendation:** None.

Rule 7.2.2: Ensure `/etc/shadow` password fields are not empty

- **Status:** PASS
- **Evidence:** `root_shadow_password_status` is `*` and system accounts have `passwd_status: "L"`, indicating no empty password fields.
- **Recommendation:** None.

Rule 7.2.3: Ensure all groups in `/etc/passwd` exist in `/etc/group`

- **Status:** PASS
- **Evidence:** `orphan_gids_in_passwd` is empty.
- **Recommendation:** None.

Rule 7.2.4: Ensure shadow group is empty

- **Status:** PASS
- **Evidence:** `shadow_group_members` is empty.
- **Recommendation:** None.

Rule 7.2.5: Ensure no duplicate UIDs exist

- **Status:** PASS
- **Evidence:** `duplicate_uids` is empty.
- **Recommendation:** None.

Rule 7.2.6: Ensure no duplicate GIDs exist

- **Status:** PASS
- **Evidence:** `duplicate_gids` is empty.
- **Recommendation:** None.

Rule 7.2.7: Ensure no duplicate user names exist

- **Status:** PASS
- **Evidence:** `duplicate_usernames` is empty.

- **Recommendation:** None.

Rule 7.2.8: Ensure no duplicate group names exist

- **Status:** PASS
- **Evidence:** `duplicate_groupnames` is empty.
- **Recommendation:** None.

Rule 7.2.9: Ensure local interactive user home directories are configured

- **Status:** PASS
- **Evidence:** `home_directory_issues` is empty.
- **Recommendation:** None.

Rule 7.2.10: Ensure local interactive user dot files access is configured

- **Status:** PASS
- **Evidence:** `dotfile_permission_issues` is empty.
- **Recommendation:** None.

SECTION 1.1 – FILESYSTEM KERNEL MODULES & PARTITIONS

Rule 1.1.1.1: Ensure cramfs filesystem is disabled

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `filesystem` data could not be collected. Ensure `cramfs` is not loaded and is blacklisted/disabled via `modprobe`.

Rule 1.1.1.2: Ensure freevxfs filesystem is disabled

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `filesystem` data could not be collected. Ensure `freevxfs` is not loaded and is blacklisted/disabled via `modprobe`.

Rule 1.1.1.3: Ensure hfs filesystem is disabled

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `filesystem` data could not be collected. Ensure `hfs` is not loaded and is blacklisted/disabled via `modprobe`.

Rule 1.1.1.4: Ensure hfsplus filesystem is disabled

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile reported an error: No JSON

object found in output: ''.

- **Recommendation:** Investigate why `filesystem` data could not be collected. Ensure `hfsplus` is not loaded and is blacklisted/disabled via `modprobe`.

Rule 1.1.1.5: Ensure jffs2 filesystem is disabled

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `filesystem` data could not be collected. Ensure `jffs2` is not loaded and is blacklisted/disabled via `modprobe`.

Rule 1.1.1.6: Ensure overlay filesystem is disabled

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `filesystem` data could not be collected. Ensure `overlay` is not loaded and is blacklisted/disabled via `modprobe`.

Rule 1.1.1.7: Ensure squashfs filesystem is disabled

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `filesystem` data could not be collected. Ensure `squashfs` is not loaded and is blacklisted/disabled via `modprobe`.

Rule 1.1.1.8: Ensure udf filesystem is disabled

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `filesystem` data could not be collected. Ensure `udf` is not loaded and is blacklisted/disabled via `modprobe`.

Rule 1.1.1.9: Ensure firewire-core module is disabled

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `filesystem` data could not be collected. Ensure `firewire-core` is not loaded and is blacklisted/disabled via `modprobe`.

Rule 1.1.1.10: Ensure usb-storage module is disabled

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `filesystem` data could not be collected. Ensure `usb-storage` is not loaded and is blacklisted/disabled via `modprobe`.

Rule 1.1.1.11: Unused filesystems

- **Status:** INFORMATIONAL
- **Evidence:** The `filesystem` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `filesystem` data could not be collected. Manually review `lsmod` output to identify and disable any unused filesystem kernel modules.

Rule 1.1.2.1.1: Ensure /tmp is a separate partition

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `filesystem` data could not be collected. Ensure `/tmp` is configured as a separate partition or a `tmpfs` filesystem.

Rule 1.1.2.1.2: Ensure nodev option is set on /tmp partition

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `filesystem` data could not be collected. Ensure the `nodev` option is set for the `/tmp` partition in `/etc/fstab`.

Rule 1.1.2.1.3: Ensure nosuid option is set on /tmp partition

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `filesystem` data could not be collected. Ensure the `nosuid` option is set for the `/tmp` partition in `/etc/fstab`.

Rule 1.1.2.1.4: Ensure noexec option is set on /tmp partition

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `filesystem` data could not be collected. Ensure the `noexec` option is set for the `/tmp` partition in `/etc/fstab`.

Rule 1.1.2.2.1: Ensure /dev/shm is a separate partition

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `filesystem` data could not be collected. Ensure `/dev/shm` is configured as a separate partition or a `tmpfs` filesystem.

Rule 1.1.2.2.2: Ensure nodev option is set on /dev/shm partition

- **Status:** UNKNOWN

- **Evidence:** The filesystem object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why filesystem data could not be collected. Ensure the nodev option is set for the /dev/shm partition in /etc/fstab.

Rule 1.1.2.2.3: Ensure nosuid option is set on /dev/shm partition

- **Status:** UNKNOWN
- **Evidence:** The filesystem object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why filesystem data could not be collected. Ensure the nosuid option is set for the /dev/shm partition in /etc/fstab.

Rule 1.1.2.2.4: Ensure noexec option is set on /dev/shm partition

- **Status:** UNKNOWN
- **Evidence:** The filesystem object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why filesystem data could not be collected. Ensure the noexec option is set for the /dev/shm partition in /etc/fstab.

Rule 1.1.2.3.1: Ensure /home is a separate partition

- **Status:** UNKNOWN
- **Evidence:** The filesystem object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why filesystem data could not be collected. Ensure /home is configured as a separate partition.

Rule 1.1.2.3.2: Ensure nodev option is set on /home partition

- **Status:** UNKNOWN
- **Evidence:** The filesystem object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why filesystem data could not be collected. Ensure the nodev option is set for the /home partition in /etc/fstab.

Rule 1.1.2.3.3: Ensure nosuid option is set on /home partition

- **Status:** UNKNOWN
- **Evidence:** The filesystem object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why filesystem data could not be collected. Ensure the nosuid option is set for the /home partition in /etc/fstab.

Rule 1.1.2.4.1: Ensure /var is a separate partition

- **Status:** UNKNOWN
- **Evidence:** The filesystem object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why filesystem data could not be collected. Ensure /var is configured as a separate partition.

Rule 1.1.2.4.2: Ensure nodev option is set on /var partition

- **Status:** UNKNOWN
- **Evidence:** The filesystem object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why filesystem data could not be collected. Ensure the nodev option is set for the /var partition in /etc/fstab.

Rule 1.1.2.4.3: Ensure nosuid option is set on /var partition

- **Status:** UNKNOWN
- **Evidence:** The filesystem object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why filesystem data could not be collected. Ensure the nosuid option is set for the /var partition in /etc/fstab.

Rule 1.1.2.5.1: Ensure /var/tmp is a separate partition

- **Status:** UNKNOWN
- **Evidence:** The filesystem object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why filesystem data could not be collected. Ensure /var/tmp is configured as a separate partition.

Rule 1.1.2.5.2: Ensure nodev option is set on /var/tmp partition

- **Status:** UNKNOWN
- **Evidence:** The filesystem object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why filesystem data could not be collected. Ensure the nodev option is set for the /var/tmp partition in /etc/fstab.

Rule 1.1.2.5.3: Ensure nosuid option is set on /var/tmp partition

- **Status:** UNKNOWN
- **Evidence:** The filesystem object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why filesystem data could not be collected. Ensure the nosuid option is set for the /var/tmp partition in /etc/fstab.

Rule 1.1.2.5.4: Ensure noexec option is set on /var/tmp partition

- **Status:** UNKNOWN
- **Evidence:** The filesystem object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why filesystem data could not be collected. Ensure the noexec option is set for the /var/tmp partition in /etc/fstab.

Rule 1.1.2.6.1: Ensure /var/log is a separate partition

- **Status:** UNKNOWN
- **Evidence:** The filesystem object in the security profile reported an error: No JSON

object found in output: ''.

- **Recommendation:** Investigate why filesystem data could not be collected. Ensure /var/log is configured as a separate partition.

Rule 1.1.2.6.2: Ensure nodev option is set on /var/log partition

- **Status:** UNKNOWN
- **Evidence:** The filesystem object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why filesystem data could not be collected. Ensure the nodev option is set for the /var/log partition in /etc/fstab.

Rule 1.1.2.6.3: Ensure nosuid option is set on /var/log partition

- **Status:** UNKNOWN
- **Evidence:** The filesystem object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why filesystem data could not be collected. Ensure the nosuid option is set for the /var/log partition in /etc/fstab.

Rule 1.1.2.6.4: Ensure noexec option is set on /var/log partition

- **Status:** UNKNOWN
- **Evidence:** The filesystem object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why filesystem data could not be collected. Ensure the noexec option is set for the /var/log partition in /etc/fstab.

Rule 1.1.2.7.1: Ensure /var/log/audit is a separate partition

- **Status:** UNKNOWN
- **Evidence:** The filesystem object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why filesystem data could not be collected. Ensure /var/log/audit is configured as a separate partition.

Rule 1.1.2.7.2: Ensure nodev option is set on /var/log/audit partition

- **Status:** UNKNOWN
- **Evidence:** The filesystem object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why filesystem data could not be collected. Ensure the nodev option is set for the /var/log/audit partition in /etc/fstab.

Rule 1.1.2.7.3: Ensure nosuid option is set on /var/log/audit partition

- **Status:** UNKNOWN
- **Evidence:** The filesystem object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why filesystem data could not be collected. Ensure the nosuid option is set for the /var/log/audit partition in /etc/fstab.

Rule 1.1.2.7.4: Ensure noexec option is set on /var/log/audit partition

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `filesystem` data could not be collected. Ensure the `noexec` option is set for the `/var/log/audit` partition in `/etc/fstab`.

SECTION 1.2 – PACKAGE MANAGEMENT

Rule 1.2.1.1: Signed-By option

- **Status:** INFORMATIONAL
- **Evidence:** The `package_management` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `package_management` data could not be collected. Manually review `APT.sources.list` files to ensure all repositories use the `Signed-By` option.

Rule 1.2.1.2: Weak dependencies

- **Status:** UNKNOWN
- **Evidence:** The `package_management` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `package_management` data could not be collected. Ensure `APT::Install-Recommends "false";` and `APT::Install-Suggests "false";` are configured in `/etc/apt/apt.conf.d/`.

Rule 1.2.1.3: GPG key files access

- **Status:** UNKNOWN
- **Evidence:** The `package_management` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `package_management` data could not be collected. Ensure GPG key files have permissions `0644` or less restrictive, and are owned by `root:root`.

Rule 1.2.1.4: /etc/apt/trusted.gpg.d dir

- **Status:** UNKNOWN
- **Evidence:** The `package_management` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `package_management` data could not be collected. Ensure `/etc/apt/trusted.gpg.d` has permissions `0755` or less restrictive, and is owned by `root:root`.

Rule 1.2.1.5: /etc/apt/auth.conf.d dir

- **Status:** UNKNOWN
- **Evidence:** The `package_management` object in the security profile reported an error: No JSON object found in output: ''.

- **Recommendation:** Investigate why `package_management` data could not be collected. Ensure `/etc/apt/auth.conf.d` does not exist or has permissions `0700` or less restrictive, and is owned by `root:root`.

Rule 1.2.1.6: files in `auth.conf.d`

- **Status:** UNKNOWN
- **Evidence:** The `package_management` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `package_management` data could not be collected. Ensure files in `/etc/apt/auth.conf.d` have permissions `0600` or less restrictive, and are owned by `root:root`.

Rule 1.2.1.7: `/usr/share/keyrings` dir

- **Status:** UNKNOWN
- **Evidence:** The `package_management` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `package_management` data could not be collected. Ensure `/usr/share/keyrings` has permissions `0755` or less restrictive, and is owned by `root:root`.

Rule 1.2.1.8: `/etc/apt/sources.list.d` dir

- **Status:** UNKNOWN
- **Evidence:** The `package_management` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `package_management` data could not be collected. Ensure `/etc/apt/sources.list.d` has permissions `0755` or less restrictive, and is owned by `root:root`.

Rule 1.2.1.9: files in `sources.list.d`

- **Status:** UNKNOWN
- **Evidence:** The `package_management` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `package_management` data could not be collected. Ensure files in `/etc/apt/sources.list.d` have permissions `0644` or less restrictive, and are owned by `root:root`.

Rule 1.2.2.1: Package updates installed

- **Status:** INFORMATIONAL
- **Evidence:** The `package_management` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `package_management` data could not be collected. Manually check for available package updates using `sudo apt update && sudo apt list --upgradable`.

SECTION 1.3 – APPARMOR

Rule 1.3.1.1: AppArmor packages installed

- **Status:** UNKNOWN
- **Evidence:** The apparmor object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why apparmor data could not be collected. Ensure apparmor and apparmor-utils packages are installed.

Rule 1.3.1.2: AppArmor enabled

- **Status:** UNKNOWN
- **Evidence:** The apparmor object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why apparmor data could not be collected. Ensure AppArmor is enabled in the kernel (via GRUB) and its service is active.

Rule 1.3.1.3: All AppArmor profiles enforcing

- **Status:** UNKNOWN
- **Evidence:** The apparmor object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why apparmor data could not be collected. Ensure all AppArmor profiles are in enforcing mode using `sudo aa-status`.

Rule 1.3.1.4: apparmor_restrict_unprivileged_unconfined

- **Status:** UNKNOWN
- **Evidence:** The apparmor object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why apparmor data could not be collected. Ensure `apparmor_restrict_unprivileged_unconfined` is set to 1 at runtime and persisted via `sysctl`.

SECTION 1.4 – BOOTLOADER

Rule 1.4.1: Bootloader password

- **Status:** UNKNOWN
- **Evidence:** The bootloader object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why bootloader data could not be collected. Configure a password for the bootloader (GRUB) to prevent unauthorized boot parameter modification.

Rule 1.4.2: Bootloader permissions

- **Status:** UNKNOWN
- **Evidence:** The bootloader object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why bootloader data could not be collected. Ensure GRUB configuration files (e.g., `/boot/grub/grub.cfg`) have permissions 0400 or

0600 and are owned by root:root.

SECTION 1.5 – PROCESS HARDENING

Rule 1.5.1: fs.protected_hardlinks

- **Status:** UNKNOWN
- **Evidence:** The process_hardening object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why process_hardening data could not be collected. Ensure fs.protected_hardlinks is set to 1 at runtime and persisted via sysctl.

Rule 1.5.2: fs.protected_symlinks

- **Status:** UNKNOWN
- **Evidence:** The process_hardening object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why process_hardening data could not be collected. Ensure fs.protected_symlinks is set to 1 at runtime and persisted via sysctl.

Rule 1.5.3: kernel.yama.ptrace_scope

- **Status:** UNKNOWN
- **Evidence:** The process_hardening object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why process_hardening data could not be collected. Ensure kernel.yama.ptrace_scope is set to 1 (or higher) at runtime and persisted via sysctl.

Rule 1.5.4: fs.suid_dumpable

- **Status:** UNKNOWN
- **Evidence:** The process_hardening object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why process_hardening data could not be collected. Ensure fs.suid_dumpable is set to 0 at runtime and persisted via sysctl.

Rule 1.5.5: kernel.dmesg_restrict

- **Status:** UNKNOWN
- **Evidence:** The process_hardening object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why process_hardening data could not be collected. Ensure kernel.dmesg_restrict is set to 1 at runtime and persisted via sysctl.

Rule 1.5.6: Prelink not installed

- **Status:** UNKNOWN
- **Evidence:** The process_hardening object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why process_hardening data could not be collected.

Ensure the `prelink` package is not installed.

Rule 1.5.7: Appport disabled

- **Status:** UNKNOWN
- **Evidence:** The `process_hardening` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `process_hardening` data could not be collected. Ensure `appport` is not installed or is disabled.

Rule 1.5.8: `kernel.kptr_restrict`

- **Status:** UNKNOWN
- **Evidence:** The `process_hardening` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `process_hardening` data could not be collected. Ensure `kernel.kptr_restrict` is set to 2 (or 1) at runtime and persisted via `sysctl`.

Rule 1.5.9: `kernel.randomize_va_space`

- **Status:** UNKNOWN
- **Evidence:** The `process_hardening` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `process_hardening` data could not be collected. Ensure `kernel.randomize_va_space` is set to 2 at runtime and persisted via `sysctl`.

Rule 1.5.10: Core dumps restricted

- **Status:** UNKNOWN
- **Evidence:** The `process_hardening` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `process_hardening` data could not be collected. Ensure core dumps are restricted by setting `ulimit -H -c 0` and `hard core 0` in `/etc/security/limits.conf`.

Rule 1.5.11: `systemd-coredump ProcessSizeMax`

- **Status:** UNKNOWN
- **Evidence:** The `process_hardening` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `process_hardening` data could not be collected. Ensure `ProcessSizeMax=0` is configured in `/etc/systemd/coredump.conf` or a drop-in file.

Rule 1.5.12: `systemd-coredump Storage`

- **Status:** UNKNOWN
- **Evidence:** The `process_hardening` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `process_hardening` data could not be collected.

Ensure `Storage=none` is configured in `/etc/systemd/coredump.conf` or a drop-in file.

SECTION 1.6 – WARNING BANNERS

Rule 1.6.1: `/etc/motd` configured

- **Status:** UNKNOWN
- **Evidence:** The `warning_banners` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `warning_banners` data could not be collected. Configure `/etc/motd` with a compliant warning banner that does not contain OS/kernel release information.

Rule 1.6.2: `/etc/issue` configured

- **Status:** UNKNOWN
- **Evidence:** The `warning_banners` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `warning_banners` data could not be collected. Configure `/etc/issue` with a compliant warning banner that does not contain OS/kernel release information.

Rule 1.6.3: `/etc/issue.net` configured

- **Status:** UNKNOWN
- **Evidence:** The `warning_banners` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `warning_banners` data could not be collected. Configure `/etc/issue.net` with a compliant warning banner that does not contain OS/kernel release information.

Rule 1.6.4: `pam_motd` configured

- **Status:** UNKNOWN
- **Evidence:** The `warning_banners` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `warning_banners` data could not be collected. Ensure `pam_motd` is configured to display the warning banner and dynamic motd scripts are disabled or reviewed.

Rule 1.6.6: access to `/etc/motd`

- **Status:** UNKNOWN
- **Evidence:** The `warning_banners` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `warning_banners` data could not be collected. Ensure `/etc/motd` has permissions `0644` or less restrictive, and is owned by `root:root`.

Rule 1.6.7: access to `/etc/issue`

- **Status:** UNKNOWN
- **Evidence:** The warning_banners object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why warning_banners data could not be collected. Ensure /etc/issue has permissions 0644 or less restrictive, and is owned by root:root.

Rule 1.6.8: access to /etc/issue.net

- **Status:** UNKNOWN
- **Evidence:** The warning_banners object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why warning_banners data could not be collected. Ensure /etc/issue.net has permissions 0644 or less restrictive, and is owned by root:root.

Rule 1.6.9: access to pam_motd files

- **Status:** UNKNOWN
- **Evidence:** The warning_banners object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why warning_banners data could not be collected. Ensure pam_motd related files (e.g., /etc/update-motd.d/*) have permissions 0644 (or 0755 for scripts) or less restrictive, and are owned by root:root.

SECTION 1.7 – GNOME DISPLAY MANAGER

Rule 1.7.1: GDM login banner

- **Status:** UNKNOWN
- **Evidence:** The gnome object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why gnome data could not be collected. If GDM is installed, ensure banner-message-enable=true and banner-message-text is configured in GDM settings.

Rule 1.7.2: disable-user-list

- **Status:** UNKNOWN
- **Evidence:** The gnome object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why gnome data could not be collected. If GDM is installed, ensure disable-user-list=true is configured in GDM settings.

Rule 1.7.3: GDM screen lock

- **Status:** UNKNOWN
- **Evidence:** The gnome object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why gnome data could not be collected. If GDM is installed, ensure lock-enabled=true and lock-delay is set to 900 seconds or less

in GDM settings.

Rule 1.7.4: GDM automount

- **Status:** UNKNOWN
- **Evidence:** The `gnome` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `gnome` data could not be collected. If GDM is installed, ensure `automount=false` and `automount-open=false` are configured in GDM settings.

Rule 1.7.5: GDM autorun-never

- **Status:** UNKNOWN
- **Evidence:** The `gnome` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `gnome` data could not be collected. If GDM is installed, ensure `autorun-never=true` is configured in GDM settings.

Rule 1.7.6: XDMCP disabled

- **Status:** UNKNOWN
- **Evidence:** The `gnome` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `gnome` data could not be collected. If GDM is installed, ensure XDMCP is disabled by verifying `Enable=true` is not present under `[xdmcp]` in GDM configuration.

Rule 1.7.7: Xwayland configured

- **Status:** UNKNOWN
- **Evidence:** The `gnome` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `gnome` data could not be collected. If GDM is installed, ensure Xwayland is configured securely (e.g., disabled if not needed) in GDM configuration.

SECTION 2 – SERVICES

Rule 2.1.1: Ensure autofs is not in use

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `services` data could not be collected. Ensure `autofs` packages are not installed and its service is not active.

Rule 2.1.2: Ensure Mail Transfer Agent is configured for local-only mode

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile reported an error: No JSON

object found in output: ''.

- **Recommendation:** Investigate why `services` data could not be collected. If an MTA is installed, ensure it is configured to listen only on the loopback interface.

Rule 2.1.3: Ensure avahi server is not in use

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `services` data could not be collected. Ensure `avahi-daemon` packages are not installed and its service is not active.

Rule 2.1.4: Only approved services listening (Manual)

- **Status:** INFORMATIONAL
- **Evidence:** The `services` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `services` data could not be collected. Manually review all listening network services using `ss -tuln` or `netstat -tuln` and ensure only approved services are active.

Rule 2.1.5: Ensure DHCP server is not in use

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `services` data could not be collected. Ensure DHCP server packages are not installed and its service is not active.

Rule 2.1.6: Ensure web server is not in use

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `services` data could not be collected. Ensure no web server packages (e.g., `apache2`, `nginx`) are installed and their services are not active.

Rule 2.1.7: Ensure DNS server (bind9) is not in use

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `services` data could not be collected. Ensure `bind9` packages are not installed and its service is not active.

Rule 2.1.9: Ensure dnsmasq is not in use

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `services` data could not be collected. Ensure

dnsmasq packages are not installed and its service is not active.

Rule 2.1.10: Ensure LDAP server (slapd) is not in use

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `services` data could not be collected. Ensure slapd packages are not installed and its service is not active.

Rule 2.1.11: Ensure IMAP and POP3 server is not in use

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `services` data could not be collected. Ensure IMAP/POP3 server packages are not installed and their services are not active.

Rule 2.1.12: Ensure NFS server is not in use

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `services` data could not be collected. Ensure NFS server packages are not installed and its service is not active.

Rule 2.1.13: Ensure NIS server (ypserv) is not in use

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `services` data could not be collected. Ensure ypserv packages are not installed and its service is not active.

Rule 2.1.14: Ensure CUPS printing service is not in use

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `services` data could not be collected. Ensure cups packages are not installed and its service is not active.

Rule 2.1.15: Ensure rpcbind is not in use

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `services` data could not be collected. Ensure rpcbind packages are not installed and its service is not active.

Rule 2.1.16: Ensure rsync daemon is not in use

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `services` data could not be collected. Ensure `rsync` daemon packages are not installed and its service is not active.

Rule 2.1.17: Ensure Samba (smbd) is not in use

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `services` data could not be collected. Ensure `samba` packages are not installed and its service is not active.

Rule 2.1.18: Ensure SNMP server (snmpd) is not in use

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `services` data could not be collected. Ensure `snmpd` packages are not installed and its service is not active.

Rule 2.1.21: Ensure Squid proxy is not in use

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `services` data could not be collected. Ensure `squid` packages are not installed and its service is not active.

Rule 2.1.22: Ensure xinetd is not in use

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `services` data could not be collected. Ensure `xinetd` packages are not installed and its service is not active.

Rule 2.1.23: Ensure X Window System is not in use

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `services` data could not be collected. Ensure X Window System packages are not installed and no Xorg processes are running.

Rule 2.2.1: Ensure NIS client (ypbind) is not installed

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `services` data could not be collected. Ensure `nis`

or ypbind client packages are not installed.

Rule 2.2.2: Ensure rsh-client is not installed

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `services` data could not be collected. Ensure `rsh-client` packages are not installed.

Rule 2.2.3: Ensure talk client is not installed

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `services` data could not be collected. Ensure `talk` client packages are not installed.

Rule 2.2.5: Ensure LDAP client (ldap-utils) is not installed

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `services` data could not be collected. Ensure `ldap-utils` packages are not installed.

Rule 2.3.1.1: Ensure a single time synchronization daemon is in use

- **Status:** UNKNOWN
- **Evidence:** The `time_sync` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `time_sync` data could not be collected. Ensure only one time synchronization daemon (e.g., `systemd-timesyncd` or `chrony`) is installed and active.

Rule 2.3.2.1: Ensure systemd-timesyncd timeserver is configured

- **Status:** UNKNOWN
- **Evidence:** The `time_sync` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `time_sync` data could not be collected. If `systemd-timesyncd` is used, ensure `NTP=` or `FallbackNTP=` is configured in `/etc/systemd/timesyncd.conf`.

Rule 2.3.2.2: Ensure systemd-timesyncd is enabled and running

- **Status:** UNKNOWN
- **Evidence:** The `time_sync` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `time_sync` data could not be collected. Ensure `systemd-timesyncd` service is enabled and active.

Rule 2.3.3.1: Ensure chrony timeserver is configured

- **Status:** UNKNOWN
- **Evidence:** The `time_sync` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `time_sync` data could not be collected. If `chrony` is used, ensure `server` or `pool` directives are configured in `/etc/chrony/chrony.conf`.

Rule 2.3.3.2: Ensure chrony process user

- **Status:** UNKNOWN
- **Evidence:** The `time_sync` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `time_sync` data could not be collected. If `chrony` is used, ensure it runs as the `_chrony` user.

Rule 2.3.3.3: Ensure chrony is enabled and running

- **Status:** UNKNOWN
- **Evidence:** The `time_sync` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `time_sync` data could not be collected. Ensure `chrony` service is enabled and active.

Rule 2.4.1.1: Ensure cron daemon is enabled and active

- **Status:** UNKNOWN
- **Evidence:** The `job_schedulers` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `job_schedulers` data could not be collected. Ensure `cron` daemon is installed, enabled, and active.

Rule 2.4.1.2: Ensure access to /etc/crontab

- **Status:** UNKNOWN
- **Evidence:** The `job_schedulers` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `job_schedulers` data could not be collected. Ensure `/etc/crontab` has permissions `0600` or `0640` and is owned by `root:root`.

Rule 2.4.1.3: Ensure access to /etc/cron.hourly

- **Status:** UNKNOWN
- **Evidence:** The `job_schedulers` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `job_schedulers` data could not be collected. Ensure `/etc/cron.hourly` has permissions `0700` or `0750` and is owned by `root:root`.

Rule 2.4.1.4: Ensure access to /etc/cron.daily

- **Status:** UNKNOWN
- **Evidence:** The `job_schedulers` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `job_schedulers` data could not be collected. Ensure `/etc/cron.daily` has permissions 0700 or 0750 and is owned by `root:root`.

Rule 2.4.1.5: Ensure access to /etc/cron.weekly

- **Status:** UNKNOWN
- **Evidence:** The `job_schedulers` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `job_schedulers` data could not be collected. Ensure `/etc/cron.weekly` has permissions 0700 or 0750 and is owned by `root:root`.

Rule 2.4.1.6: Ensure access to /etc/cron.monthly

- **Status:** UNKNOWN
- **Evidence:** The `job_schedulers` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `job_schedulers` data could not be collected. Ensure `/etc/cron.monthly` has permissions 0700 or 0750 and is owned by `root:root`.

Rule 2.4.1.7: Ensure access to /etc/cron.yearly

- **Status:** UNKNOWN
- **Evidence:** The `job_schedulers` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `job_schedulers` data could not be collected. Ensure `/etc/cron.yearly` has permissions 0700 or 0750 and is owned by `root:root`.

Rule 2.4.1.8: Ensure access to /etc/cron.d

- **Status:** UNKNOWN
- **Evidence:** The `job_schedulers` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `job_schedulers` data could not be collected. Ensure `/etc/cron.d` has permissions 0700 or 0750 and is owned by `root:root`.

Rule 2.4.1.9: Ensure access to crontab mechanism

- **Status:** UNKNOWN
- **Evidence:** The `job_schedulers` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `job_schedulers` data could not be collected. Ensure `/var/spool/cron/crontabs` has permissions 1730 or 0700/0755 and `crontab` binary is owned by `root`.

Rule 2.4.2.1: Ensure access to at configured

- **Status:** UNKNOWN
- **Evidence:** The `job_schedulers` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `job_schedulers` data could not be collected. Ensure `at.allow` exists with restrictive permissions and `at.deny` does not exist or is empty.

SECTION 3 – NETWORK CONFIGURATION

Rule 3.1.1: IPv6 status identified

- **Status:** INFORMATIONAL
- **Evidence:** The `network_config` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `network_config` data could not be collected. Manually identify the IPv6 status (enabled/disabled) on the system.

Rule 3.1.2: Ensure wireless interfaces are not available

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `network_config` data could not be collected. Ensure no wireless interfaces are detected and no wireless kernel modules are loaded.

Rule 3.1.3: Ensure Bluetooth services are not in use

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `network_config` data could not be collected. Ensure `bluez` is not installed or its service is disabled and inactive.

Rule 3.2.1: Ensure atm kernel module is disabled

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `network_config` data could not be collected. Ensure `atm` is not loaded and is blacklisted/disabled via `modprobe`.

Rule 3.2.2: Ensure can kernel module is disabled

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `network_config` data could not be collected. Ensure `can` is not loaded and is blacklisted/disabled via `modprobe`.

Rule 3.2.3: Ensure dccp kernel module is disabled

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `network_config` data could not be collected. Ensure `dccp` is not loaded and is blacklisted/disabled via `modprobe`.

Rule 3.2.4: Ensure `rds` kernel module is disabled

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `network_config` data could not be collected. Ensure `rds` is not loaded and is blacklisted/disabled via `modprobe`.

Rule 3.2.5: Ensure `sctp` kernel module is disabled

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `network_config` data could not be collected. Ensure `sctp` is not loaded and is blacklisted/disabled via `modprobe`.

Rule 3.2.6: Ensure `tipc` kernel module is disabled

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `network_config` data could not be collected. Ensure `tipc` is not loaded and is blacklisted/disabled via `modprobe`.

Rule 3.3.1.1: Ensure `net.ipv4.ip_forward` is configured

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `network_config` data could not be collected. Ensure `net.ipv4.ip_forward` is set to 0 at runtime and persisted via `sysctl`.

Rule 3.3.1.2: Ensure `net.ipv4.conf.all.forwarding` is configured

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `network_config` data could not be collected. Ensure `net.ipv4.conf.all.forwarding` is set to 0 at runtime and persisted via `sysctl`.

Rule 3.3.1.3: Ensure `net.ipv4.conf.default.forwarding` is configured

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile reported an error: No JSON object found in output: ''.

- **Recommendation:** Investigate why `network_config` data could not be collected. Ensure `net.ipv4.conf.default.forwarding` is set to 0 at runtime and persisted via `sysctl`.

Rule 3.3.1.4: Ensure `net.ipv4.conf.all.send_redirects` is configured

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `network_config` data could not be collected. Ensure `net.ipv4.conf.all.send_redirects` is set to 0 at runtime and persisted via `sysctl`.

Rule 3.3.1.5: Ensure `net.ipv4.conf.default.send_redirects` is configured

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `network_config` data could not be collected. Ensure `net.ipv4.conf.default.send_redirects` is set to 0 at runtime and persisted via `sysctl`.

Rule 3.3.1.6: Ensure `net.ipv4.icmp_ignore_bogus_error_responses` is configured

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `network_config` data could not be collected. Ensure `net.ipv4.icmp_ignore_bogus_error_responses` is set to 1 at runtime and persisted via `sysctl`.

Rule 3.3.1.7: Ensure `net.ipv4.icmp_echo_ignore_broadcasts` is configured

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `network_config` data could not be collected. Ensure `net.ipv4.icmp_echo_ignore_broadcasts` is set to 1 at runtime and persisted via `sysctl`.

Rule 3.3.1.8: Ensure `net.ipv4.conf.all.accept_redirects` is configured

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `network_config` data could not be collected. Ensure `net.ipv4.conf.all.accept_redirects` is set to 0 at runtime and persisted via `sysctl`.

Rule 3.3.1.9: Ensure `net.ipv4.conf.default.accept_redirects` is configured

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `network_config` data could not be collected. Ensure `net.ipv4.conf.default.accept_redirects` is set to 0 at runtime and persisted via `sysctl`.

Rule 3.3.1.10: Ensure `net.ipv4.conf.all.secure_redirects` is configured

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `network_config` data could not be collected. Ensure `net.ipv4.conf.all.secure_redirects` is set to 0 at runtime and persisted via `sysctl`.

Rule 3.3.1.11: Ensure `net.ipv4.conf.default.secure_redirects` is configured

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `network_config` data could not be collected. Ensure `net.ipv4.conf.default.secure_redirects` is set to 0 at runtime and persisted via `sysctl`.

Rule 3.3.1.12: Ensure `net.ipv4.conf.all.rp_filter` is configured

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `network_config` data could not be collected. Ensure `net.ipv4.conf.all.rp_filter` is set to 1 or 2 at runtime and persisted via `sysctl`.

Rule 3.3.1.13: Ensure `net.ipv4.conf.default.rp_filter` is configured

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `network_config` data could not be collected. Ensure `net.ipv4.conf.default.rp_filter` is set to 1 or 2 at runtime and persisted via `sysctl`.

Rule 3.3.1.14: Ensure `net.ipv4.conf.all.accept_source_route` is configured

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `network_config` data could not be collected. Ensure `net.ipv4.conf.all.accept_source_route` is set to 0 at runtime and persisted via `sysctl`.

Rule 3.3.1.15: Ensure net.ipv4.conf.default.accept_source_route is configured

- **Status:** UNKNOWN
- **Evidence:** The network_config object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why network_config data could not be collected. Ensure net.ipv4.conf.default.accept_source_route is set to 0 at runtime and persisted via sysctl.

Rule 3.3.1.16: Ensure net.ipv4.conf.all.log_martians is configured

- **Status:** UNKNOWN
- **Evidence:** The network_config object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why network_config data could not be collected. Ensure net.ipv4.conf.all.log_martians is set to 1 at runtime and persisted via sysctl.

Rule 3.3.1.17: Ensure net.ipv4.conf.default.log_martians is configured

- **Status:** UNKNOWN
- **Evidence:** The network_config object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why network_config data could not be collected. Ensure net.ipv4.conf.default.log_martians is set to 1 at runtime and persisted via sysctl.

Rule 3.3.1.18: Ensure net.ipv4.tcp_syncookies is configured

- **Status:** UNKNOWN
- **Evidence:** The network_config object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why network_config data could not be collected. Ensure net.ipv4.tcp_syncookies is set to 1 at runtime and persisted via sysctl.

Rule 3.3.2.1: Ensure net.ipv6.conf.all.forwarding is configured

- **Status:** UNKNOWN
- **Evidence:** The network_config object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why network_config data could not be collected. Ensure net.ipv6.conf.all.forwarding is set to 0 at runtime and persisted via sysctl.

Rule 3.3.2.2: Ensure net.ipv6.conf.default.forwarding is configured

- **Status:** UNKNOWN
- **Evidence:** The network_config object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why network_config data could not be collected. Ensure net.ipv6.conf.default.forwarding is set to 0 at runtime and persisted

via `sysctl`.

Rule 3.3.2.3: Ensure `net.ipv6.conf.all.accept_redirects` is configured

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `network_config` data could not be collected. Ensure `net.ipv6.conf.all.accept_redirects` is set to 0 at runtime and persisted via `sysctl`.

Rule 3.3.2.4: Ensure `net.ipv6.conf.default.accept_redirects` is configured

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `network_config` data could not be collected. Ensure `net.ipv6.conf.default.accept_redirects` is set to 0 at runtime and persisted via `sysctl`.

Rule 3.3.2.5: Ensure `net.ipv6.conf.all.accept_source_route` is configured

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `network_config` data could not be collected. Ensure `net.ipv6.conf.all.accept_source_route` is set to 0 at runtime and persisted via `sysctl`.

Rule 3.3.2.6: Ensure `net.ipv6.conf.default.accept_source_route` is configured

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `network_config` data could not be collected. Ensure `net.ipv6.conf.default.accept_source_route` is set to 0 at runtime and persisted via `sysctl`.

Rule 3.3.2.7: Ensure `net.ipv6.conf.all.accept_ra` is configured

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile reported an error: No JSON object found in output: ''.
- **Recommendation:** Investigate why `network_config` data could not be collected. Ensure `net.ipv6.conf.all.accept_ra` is set to 0 at runtime and persisted via `sysctl`.

Rule 3.3.2.8: Ensure `net.ipv6.conf.default.accept_ra` is configured

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile reported an error: No

JSON object found in output: ''.

- **Recommendation:** Investigate why `network_config` data could not be collected. Ensure `net.ipv6.conf.default.accept_ra` is set to 0 at runtime and persisted via `sysctl`.

SECTION 5.3 – PLUGGABLE AUTHENTICATION MODULES (PAM)

Rule 5.3.1.1: Ensure `libpam-runtime` is installed and up-to-date

- **Status:** PASS
- **Evidence:** `libpam-runtime` is installed (`installed: true`) and `installed_version (1.5.3-5ubuntu5.5)` matches `candidate_version (1.5.3-5ubuntu5.5)`.
- **Recommendation:** None.

Rule 5.3.1.2: Ensure `libpam-modules` is installed and up-to-date

- **Status:** PASS
- **Evidence:** `libpam-modules` is installed (`installed: true`) and `installed_version (1.5.3-5ubuntu5.5)` matches `candidate_version (1.5.3-5ubuntu5.5)`.
- **Recommendation:** None.

Rule 5.3.1.3: Ensure `libpam-pwquality` is installed and up-to-date

- **Status:** FAIL
- **Evidence:** `libpam-pwquality` is not installed (`installed: false`).
- **Recommendation:** Install the `libpam-pwquality` package using `sudo apt install libpam-pwquality`.

Rule 5.3.1.4: Ensure `cracklib-runtime` is installed and up-to-date

- **Status:** FAIL
- **Evidence:** `cracklib-runtime` is not installed (`installed: false`).
- **Recommendation:** Install the `cracklib-runtime` package using `sudo apt install cracklib-runtime`.

Rule 5.3.2.1: Ensure `pam_unix` is enabled

- **Status:** PASS
- **Evidence:** `pam_unix.so` module lines are present and active in `common-auth`, `common-account`, `common-password`, and `common-session`.
- **Recommendation:** None.

Rule 5.3.2.2: Ensure `pam_faillock` is enabled

- **Status:** FAIL
- **Evidence:** `pam_faillock.so` module lines are not present in any common PAM configuration files.
- **Recommendation:** Configure `pam_faillock.so` in `/etc/pam.d/common-auth`

and `/etc/pam.d/common-account` to lock accounts after failed login attempts.

Rule 5.3.2.3: Ensure pam_pwquality is enabled

- **Status:** FAIL
- **Evidence:** `pam_pwquality.so` module lines are not present in any common PAM configuration files.
- **Recommendation:** Configure `pam_pwquality.so` in `/etc/pam.d/common-password` to enforce password quality requirements.

Rule 5.3.2.4: Ensure pam_pwhistory is enabled

- **Status:** FAIL
- **Evidence:** `pam_pwhistory.so` module lines are not present in any common PAM configuration files.
- **Recommendation:** Configure `pam_pwhistory.so` in `/etc/pam.d/common-password` to enforce password history.

Rule 5.3.3.1.1: Ensure pam_faillock deny is configured to 5 or less

- **Status:** FAIL
- **Evidence:** `pam_faillock` is not enabled, and `deny` setting is null.
- **Recommendation:** Enable `pam_faillock` and set `deny = 5` (or less) in `/etc/security/faillock.conf` or as an inline argument in `/etc/pam.d/common-auth`.

Rule 5.3.3.1.2: Ensure pam_faillock unlock_time is configured to 900 seconds or more

- **Status:** FAIL
- **Evidence:** `pam_faillock` is not enabled, and `unlock_time` setting is null.
- **Recommendation:** Enable `pam_faillock` and set `unlock_time = 900` (or 0 for never unlock) in `/etc/security/faillock.conf` or as an inline argument in `/etc/pam.d/common-auth`.

Rule 5.3.3.1.3: Ensure pam_faillock even_deny_root and root_unlock_time are configured

- **Status:** FAIL
- **Evidence:** `pam_faillock` is not enabled, and `even_deny_root` and `root_unlock_time` settings are null.
- **Recommendation:** Enable `pam_faillock` and set `even_deny_root` and `root_unlock_time = 60` (or more) in `/etc/security/faillock.conf` or as an inline argument in `/etc/pam.d/common-auth`.

Rule 5.3.3.2.1: Ensure pam_pwquality difok is configured to 2 or more

- **Status:** FAIL
- **Evidence:** `pam_pwquality` is not enabled, and `difok` setting is null.
- **Recommendation:** Enable `pam_pwquality` and set `difok = 2` (or more) in `/etc/security/pwquality.conf` or as an inline argument in `/etc/pam.d/common-password`.

Rule 5.3.3.2.2: Ensure pam_pwquality minlen is configured to 14 or more

- **Status:** FAIL
- **Evidence:** `pam_pwquality` is not enabled, and `minlen` setting is null.
- **Recommendation:** Enable `pam_pwquality` and set `minlen = 14` (or more) in `/etc/security/pwquality.conf` or as an inline argument in `/etc/pam.d/common-password`.

Rule 5.3.3.2.3: Password complexity (dcredit, ucredit, lcredit, ocredit)

- **Status:** INFORMATIONAL
- **Evidence:** `dcredit`, `ucredit`, `lcredit`, `ocredit` settings are null, indicating default values are used or `pam_pwquality` is not enabled.
- **Recommendation:** Manually review password complexity requirements. If `pam_pwquality` is enabled, configure `dcredit`, `ucredit`, `lcredit`, `ocredit` in `/etc/security/pwquality.conf` to enforce strong password complexity.

Rule 5.3.3.2.4: Ensure pam_pwquality maxrepeat is configured to 3 or less

- **Status:** FAIL
- **Evidence:** `pam_pwquality` is not enabled, and `maxrepeat` setting is null.
- **Recommendation:** Enable `pam_pwquality` and set `maxrepeat = 3` (or less, but not 0) in `/etc/security/pwquality.conf` or as an inline argument in `/etc/pam.d/common-password`.

Rule 5.3.3.2.5: Ensure pam_pwquality maxsequence is configured to 3 or less

- **Status:** FAIL
- **Evidence:** `pam_pwquality` is not enabled, and `maxsequence` setting is null.
- **Recommendation:** Enable `pam_pwquality` and set `maxsequence = 3` (or less, but not 0) in `/etc/security/pwquality.conf` or as an inline argument in `/etc/pam.d/common-password`.

Rule 5.3.3.2.6: Ensure pam_pwquality dictcheck is configured to 1

- **Status:** FAIL
- **Evidence:** `pam_pwquality` is not enabled, and `dictcheck` setting is null.
- **Recommendation:** Enable `pam_pwquality` and set `dictcheck = 1` in `/etc/security/pwquality.conf` or as an inline argument in `/etc/pam.d/common-password`.

Rule 5.3.3.2.7: Ensure pam_pwquality enforcing is configured to 1

- **Status:** FAIL
- **Evidence:** `pam_pwquality` is not enabled, and `enforcing` setting is null.
- **Recommendation:** Enable `pam_pwquality` and set `enforcing = 1` in `/etc/security/pwquality.conf` or as an inline argument in `/etc/pam.d/common-password`.

Rule 5.3.3.2.8: Ensure pam_pwquality enforce_for_root is present

- **Status:** FAIL
- **Evidence:** `pam_pwquality` is not enabled, and `enforce_for_root` setting is null.
- **Recommendation:** Enable `pam_pwquality` and set `enforce_for_root` in `/etc/security/pwquality.conf` or as an inline argument in `/etc/pam.d/common-password`.

Rule 5.3.3.3.1: Ensure pam_pwhistory remember is configured to 24 or more

- **Status:** FAIL
- **Evidence:** `pam_pwhistory` is not enabled, and `remember` setting is null.
- **Recommendation:** Enable `pam_pwhistory` and set `remember = 24` (or more) as an inline argument in `/etc/pam.d/common-password`.

Rule 5.3.3.3.2: Ensure pam_pwhistory enforce_for_root is present

- **Status:** FAIL
- **Evidence:** `pam_pwhistory` is not enabled, and `enforce_for_root` setting is null.
- **Recommendation:** Enable `pam_pwhistory` and add `enforce_for_root` to its inline arguments in `/etc/pam.d/common-password`.

Rule 5.3.3.3.3: Ensure pam_pwhistory use_authtok is present

- **Status:** FAIL
- **Evidence:** `pam_pwhistory` is not enabled, and `use_authtok` setting is null.
- **Recommendation:** Enable `pam_pwhistory` and add `use_authtok` to its inline arguments in `/etc/pam.d/common-password`.

Rule 5.3.3.4.1: Ensure pam_unix nullok is absent

- **Status:** FAIL
- **Evidence:** `nullok` is present in `pam_unix.so` line in `/etc/pam.d/common-auth`.
- **Recommendation:** Remove the `nullok` option from `pam_unix.so` in `/etc/pam.d/common-auth` to prevent users from logging in with empty passwords.

Rule 5.3.3.4.2: Ensure pam_unix remember is absent

- **Status:** PASS
- **Evidence:** `remember` is not present in `pam_unix.so` lines.
- **Recommendation:** None.

Rule 5.3.3.4.3: Ensure pam_unix uses strong hashing

- **Status:** PASS
- **Evidence:** `yescrypt` is present in `pam_unix.so` line in `/etc/pam.d/common-password`.
- **Recommendation:** None.

Rule 5.3.3.4.4: Ensure pam_unix use_authtok is present (in password stack)

- **Status:** FAIL

- **Evidence:** use_authtok is not present in pam_unix.so line in /etc/pam.d/common-password.
 - **Recommendation:** Add the use_authtok option to pam_unix.so in /etc/pam.d/common-password to ensure the new password is used for all stacked modules.
-

SUID/SGID Triage Report

```
{
  "findings": [],
  "summary": {
    "suspicious_found": 0,
    "message": "No unexpected SUID/SGID files detected."
  }
}
```